

Willingness to Pay for Parking and Commuter Mode Choice: A Discrete Choice Analysis with Counterfactual Simulation

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1. Data

2. Sample Representativeness

##	gender_dummy	group_dummy	child_dummy	car_primary_dummy	car_secondary_dummy
## 1	1	1	0	0	0
## 2	1	1	0	0	0
## 3	0	1	0	0	0
## 4	1	1	0	0	0
## 5	1	1	0	0	1
## 6	1	1	0	0	0
##	indv_inc_numeric	hh_inc_numeric			
## 1	1	1			
## 2	1	1			
## 3	0	0			
## 4	1	0			
## 5	1	2			
## 6	1	0			

Table 1: Sample Summary Statistics

Share Female	Share Students	Share with Child	Share Car (Primary)	Share Car (Secondary)	Mean Age
0.678	0.778	0.119	0.161	0.215	28.835

The sample seems to be younger, more female, and more student-heavy than the general population. In particular, 67.8% of respondents are female, and 77.8% are students, both being much greater than typical share given the general population characteristics. The sample has a mean age of 28.84 years, which signals a younger population overall, and just 11.9% of respondents have children, indicating that older adults and

parents are likely underrepresented. Apart from that, 16.1% of respondents report having a car as their main mode and 21.5% as a secondary mode, which may be lower than population averages and may reflect that a more urban or campus-based sample does not rely on this mode of transportation as much as alternate modes of transport.

The age distribution plot shows that the majority of respondents are in their 20s, with a particularly high concentration between ages 20 and 30, especially among females. Males also cluster in the younger age range but with slightly more spread toward the 30–40 range. Very few respondents are over 50, indicating that older age groups are underrepresented. The “Other” gender category has very few observations overall, spread sparsely across ages.

The transportation plot reveals that bike or e-bike is the most common primary transportation mode for both genders, but especially for females, who dominate this category. Males are more likely than females to use cars as their primary transport, while females choose buses and trains more frequently than males. Walking is also more common among females. Motor scooter or motorbike usage is minimal for both genders. This distribution suggests a sample with lower car dependency than might be expected in a general population, potentially reflecting an urban or student-oriented environment.

3. Willingness to Pay for Parking

Table 2: WTP Summary by Randomisation Group

Group	N	Mean WTP (€/h)	Median WTP (€/h)	SD
A (price only)	43	0.256	0	0.486
B (price + env. info)	22	0.284	0	0.619
C (price + cost info)	27	0.148	0	0.296

The average weekly willingness to pay for parking was **€0.25** for respondents exposed only to basic price information (group A = 1). Group B (=2) was shown some environmental information (environmentally friendly car usage), their average price was **€0.28** for parking. Group C (=3) had a lower average price of **€0.15**, showing costs of car usage. Clearly, in all three groups the average reservation prices were quite low. Most respondents refused to pay any fee at all for a parking spot, which resulted in many zero WTP responses and a highly skewed distribution overall. Because of these low WTP observations, differences were relatively small due to lowered levels of overall willingness to pay across groups.

4. OLS Regression — WTP Determinants

Across all specifications the explanatory power is very low ($R^2 = 0.069, 0.118, 0.069, 0.081, 0.097$), reflecting the substantial unexplained heterogeneity in WTP. Gender and student status enter with the expected signs—males and students tend to accept slightly higher hourly parking fees—but the coefficients are not statistically significant at conventional levels. Income variables have the anticipated positive influence: individuals with higher personal incomes are willing to pay more for parking. Age does not play a major role in explaining WTP. When the dummies are added (Model 5) the fit improves only marginally; group B exhibits a modest positive effect relative to group A, whereas group C’s effect is small and negative. However, both of the treatment dummies are not statistically significant in model 5.

Model 5 is still preferred with the second biggest R^2 because it accounts for the randomized information, enabling an unbiased comparison across groups. Nonetheless, the model suffers from several shortcomings.

- 1) The number of potential car users $n = 92$ is small and many respondents report zero WTP, leading to little variation in the dependent variable and imprecise estimates.
- 2) The WTP questions generate hypothetical willingness to pay, and may also be strategic or protest responses that individuals simply reject paying for parking.

Table 3: OLS Regression: WTP Determinants (HC1 Robust SE)

	m1	m2	m3	m4	m5
(Intercept)	0.197*	-0.010	0.138	0.001	0.036
	(0.111)	(0.181)	(0.276)	(0.305)	(0.318)
Female	-0.100	-0.114	-0.097	-0.098	-0.098
	(0.099)	(0.098)	(0.101)	(0.101)	(0.101)
Student	0.041	0.149	0.066	0.137	0.097
	(0.109)	(0.130)	(0.152)	(0.166)	(0.171)
Has Child	0.289**	0.236*	0.283**	0.268**	0.266**
	(0.126)	(0.126)	(0.129)	(0.130)	(0.131)
Personal Income		-0.020		0.075	0.074
		(0.086)		(0.071)	(0.071)
Household Income		0.105*			
		(0.055)			
Age			0.001	0.001	0.001
			(0.006)	(0.006)	(0.006)
Group B (env. info)					0.086
					(0.125)
Group C (cost info)					-0.083
					(0.119)
Observations	92	92	92	92	92
R ²	0.069	0.118	0.069	0.081	0.097
Adj.R ²	0.037	0.067	0.027	0.028	0.022

* p < 0.1, ** p < 0.05, *** p < 0.01

- 3) Important determinants of WTP, such as current parking experience, perceived enforcement and attitudes toward environmental policies, are omitted from the regression, introducing omitted variable bias. Finally, the income categories are coarse, and misclassification could attenuate the estimated effects.

5. Conditional Logit Model — Transport Mode Choice

```
##          Bike or E-Bike          Bus
##          7.814435e-01          7.302961e-04
##          Car Motor scooter or Motorbike
##          1.369026e-01          3.449166e-166
##          Train          Walking
##          8.092356e-02          1.248705e-12
```

We now model the first transport choice using a conditional (multinomial) logit model. The dependent variable is the categorical variable `primary_transp` (choices: Car, Bike/E-Bike, Bus, Train, Walking, Motor scooter/Motorbike). Explanatory variables include the dummies defined above, age, personal income, child dummy and the distance from home to the university (HC1, in kilometres). To estimate a conditional logit in R we reshape the data into long format using the `dfidx` package.

The conditional logit results (not shown here for simplicity) indicate that **distance to the university** significantly affects mode choice: longer distances increase the probability of choosing Car or Train and reduce the probability of choosing active modes such as walking or cycling. Interactions between distance and gender or child status can be added to allow the effect of distance to differ across groups.

Our preferred specification includes distance, gender dummy, age, student status, income dummies, child dummy and interaction terms between distance and gender as well as between distance and child status. This model yields intuitive signs and improves the fit without over-fitting.

The coefficient estimates for “Motor scooter or Motorbike” are extremely large in magnitude with correspondingly large standard errors. This is due to the very low number of observations for this alternative (<1%), leading to quasi-complete separation in the logit model. While the alternative is retained in the model to preserve the integrity of the choice set, these coefficients are not meaningful for interpretation.

6. IIA Property & Gender Comparison

```
##          Pair
## Bike or E-Bike          Bike or E-Bike / Car
## Bike or E-Bike1          Bike or E-Bike / Motor scooter or Motorbike
## Bike or E-Bike2          Bike or E-Bike / Train
## Bike or E-Bike3          Bike or E-Bike / Walking
## Car          Car / Motor scooter or Motorbike
## Car1          Car / Train
## Car2          Car / Walking
## Motor scooter or Motorbike          Motor scooter or Motorbike / Train
## Motor scooter or Motorbike1          Motor scooter or Motorbike / Walking
## Train          Train / Walking
##          OR_with_bus OR_without_bus
## Bike or E-Bike          3.57142857          3.57142857
## Bike or E-Bike1          149.99998771          149.99998680
## Bike or E-Bike2          6.52173913          6.52173913
## Bike or E-Bike3          8.33333335          8.33333334
## Car          41.99999656          41.99999631
## Car1          1.82608696          1.82608696
## Car2          2.33333334          2.33333333
## Motor scooter or Motorbike          0.04347826          0.04347826
## Motor scooter or Motorbike1          0.05555556          0.05555556
```

```
## Train                                1.27777778    1.27777778
## # A tibble: 8 x 5
##   gender_dummy Scenario Transport_Mode Probability Gender
##   <dbl> <chr> <chr> <dbl> <chr>
## 1         0 With Bus Walking      0.0714 Male
## 2         0 With Bus Bike or E-Bike 0.524 Male
## 3         1 With Bus Walking      0.0678 Female
## 4         1 With Bus Bike or E-Bike 0.599 Female
## 5         0 Without Bus Walking      0.0759 Male
## 6         0 Without Bus Bike or E-Bike 0.557 Male
## 7         1 Without Bus Walking      0.0774 Female
## 8         1 Without Bus Bike or E-Bike 0.684 Female
```

The IIA property implies that the relative odds between any two alternatives remain the same even if a third option is removed. In our results, removing Bus hardly changes the odds ratios — for example, the odds of choosing Bike/E-Bike over Car stay at 3.5714 both with and without Bus. This shows the model fails to reflect realistic substitution patterns; in reality, removing Bus should shift probabilities toward certain modes more than others.

By gender, the predicted probabilities show that females are more likely than males to come by bike/e-bike or on foot. With Bus in the model, female probabilities are 0.5989 for Bike/E-Bike and 0.0678 for Walking, compared to 0.5238 and 0.0714 for males. When Bus is removed, female probabilities rise to 0.6839 (Bike/E-Bike) and 0.0774 (Walking), while male probabilities increase to 0.5570 and 0.0759, respectively.

The minimal change in these probabilities after removing Bus illustrates how the IIA property constrains the model’s responsiveness to changes in available alternatives.

7. Marginal Effects & Elasticities

In a multinomial logit model the probability that individual i chooses alternative j is

$$P_{ij} = \frac{\exp(V_{ij})}{\sum_k \exp(V_{ik})},$$

where V_{ij} is the systematic utility of alternative j . In our preferred specification the utility for the *Car* option is

$$V_{i,\text{car}} = \alpha_{\text{car}} + \beta_{\text{gender}} \cdot \text{gender}_i + \beta_{\text{age}} \cdot \text{age}_i + \beta_{\text{group}} \cdot \text{student}_i + \beta_{\text{inc}} \cdot \text{income}_i + \beta_{\text{child}} \cdot \text{child}_i + \beta_{\text{dist}} \cdot d_i,$$

with d_i denoting the distance between the place of living and the university. Analogous expressions hold for the other alternatives, each with its own intercept and slope parameters.

For a continuous explanatory variable x (e.g., distance) the marginal effect on the probability of choosing alternative j is

$$\frac{\partial P_{ij}}{\partial x_i} = P_{ij}(\beta_{j,x} - \sum_k P_{ik} \beta_{k,x}).$$

Table 4: Marginal Effects of Covariates

Covariate	Alternative	Marginal Effect
gender_dummy	Bike or E-Bike	0.1649
gender_dummy	Bus	0.0007

Covariate	Alternative	Marginal Effect
gender_dummy	Car	-0.1606
gender_dummy	Motor scooter or Motorbike	0.0000
gender_dummy	Train	-0.0051
gender_dummy	Walking	0.0000
age	Bike or E-Bike	0.0001
age	Bus	-0.0001
age	Car	0.0004
age	Motor scooter or Motorbike	0.0000
age	Train	-0.0004
age	Walking	0.0000
group_dummy	Bike or E-Bike	-0.0132
group_dummy	Bus	0.0171
group_dummy	Car	-0.0621
group_dummy	Motor scooter or Motorbike	0.0000
group_dummy	Train	0.0582
group_dummy	Walking	0.0000
indv_inc_numeric	Bike or E-Bike	0.0528
indv_inc_numeric	Bus	-0.0003
indv_inc_numeric	Car	-0.0126
indv_inc_numeric	Motor scooter or Motorbike	0.0000
indv_inc_numeric	Train	-0.0399
indv_inc_numeric	Walking	0.0000
hh_inc_numeric	Bike or E-Bike	-0.0500
hh_inc_numeric	Bus	0.0002
hh_inc_numeric	Car	0.0291
hh_inc_numeric	Motor scooter or Motorbike	0.0000
hh_inc_numeric	Train	0.0207
hh_inc_numeric	Walking	0.0000
child_dummy	Bike or E-Bike	-0.2127
child_dummy	Bus	0.0013
child_dummy	Car	0.0831
child_dummy	Motor scooter or Motorbike	0.0000
child_dummy	Train	0.1283
child_dummy	Walking	0.0000
distance	Bike or E-Bike	-0.0120
distance	Bus	0.0000
distance	Car	0.0070
distance	Motor scooter or Motorbike	0.0000
distance	Train	0.0050
distance	Walking	0.0000
distance_bus	Bike or E-Bike	-0.0174
distance_bus	Bus	0.0000
distance_bus	Car	0.0417
distance_bus	Motor scooter or Motorbike	0.0000
distance_bus	Train	-0.0243
distance_bus	Walking	0.0000
distance_train	Bike or E-Bike	-0.0126
distance_train	Bus	0.0001
distance_train	Car	0.0126
distance_train	Motor scooter or Motorbike	0.0000
distance_train	Train	-0.0001
distance_train	Walking	0.0000

Table 5: Own Elasticities with Respect to Distance

Alternative	Elasticity
Bike or E-Bike	-0.2172
Bus	-0.1762
Car	0.7236
Motor scooter or Motorbike	-217.2757
Train	0.8748
Walking	-29.2327

The estimated marginal effects indicate how changes in each explanatory variable affect the probability of choosing each transport option, holding other factors constant. For example, the marginal effect of distance on car use is 0.0070, meaning that a one-kilometer increase in distance increases the probability of choosing a car by 0.7 percentage points on average. The own-elasticity of car use with respect to distance is 0.72, implying that a 1 % increase in distance raises the probability of commuting by car by about 0.72 %. Conversely, the probability of walking decreases sharply with distance (elasticity -29.23), and the probability of cycling also falls (elasticity -0.22), indicating strong substitution away from active modes toward motorized transport as distance grows. The probability of using the train increases with distance (elasticity 0.87), suggesting complementarity in longer trips.

However, the results for the “Motor scooter or Motorbike” alternative show extremely large and implausible elasticities (-217.28). This is likely due to data imbalance, as this category accounts for less than 1 % of the sample, which makes parameter estimates unstable. Therefore, these results should be interpreted with caution and are not economically meaningful.

##	distance	distance_bus	distance_train
## Bike or E-Bike	-0.2171898	-0.02092825	-0.065786864
## Bus	-0.1761776	0.05868974	0.359578600
## Car	0.7235913	0.28580410	0.375712929
## Motor scooter or Motorbike	-217.2757191	-47.19440924	63.231978098
## Train	0.8747597	-0.28194442	-0.003583161
## Walking	-29.2327423	0.07537934	0.601896271

The variable is distance from home to the university, which is common to all modes but affects them differently via their estimated coefficients. The elasticity results indicate that increasing distance generally shifts transport choices away from short-distance modes toward those better suited for longer travel. Specifically, a 1% increase in distance reduces the probability of choosing a bike or e-bike by approximately 0.22% and decreases the likelihood of taking the bus by about 0.18%, holding other factors constant. In contrast, train usage shows a strong positive response, with a 0.88% increase in probability for every 1% increase in distance, suggesting that longer commutes encourage a shift toward rail transport. Walking exhibits an extremely large negative elasticity (-29.23), reflecting the steep decline in walking probability as distance increases and the very small baseline share of walking in the dataset. Similarly, the motor scooter or motorbike elasticity (-217.28) is unrealistically large due to the very low number of observations for this mode; small changes in probability translate into disproportionately large elasticities. These extreme values for walking and motor scooter/motorbike should be interpreted as artefacts of the dataset rather than reliable behavioural estimates.

8. Counterfactual Analysis — Student Housing Policy

To assess the effect of this relocation on transport choices, we adjust `distance` for respondents with `distance > 5` and recompute predicted probabilities using the conditional logit model.

##	BaselineCarProb	CounterfactualCarProb	Difference
## 1	0.3590441	0.1786045	-0.1804396

The baseline predicted probability of choosing the car among respondents who live beyond 5 km is approximately 36 %. When we hypothetically move those people within 5 km of their destination, the predicted probability of choosing the car drops to approximately 18 %.

This substantial difference highlights the powerful impact commuting distance has on travel mode choice. Practically, nearly one in five people who would otherwise drive, could switch to a different mode, if their commuting distance decreased. This also indicates urban planning and transportation policies can help decrease car dependence by reducing effective travel distance. Whether it facilitates student housing close to campus, encouraging mixed-use forms of development, or increasing access to services, reducing effective commuting distance acts as a powerful leverage point to reduce dependence on car travel.